Direct Prefix Counting and Deterministic Synchronization Collapse

Paper V: Separating Gate Mass from First-Record Prefix Density

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Abstract

We establish a critical distinction between two quantities that previous papers conflated:

- *Gate mass*: the fraction of all gate-1 renewal events (after a K_i record) at which $q \equiv r_{K_{i+1}} \pmod{2^{K_{i+1}}}$.
- *First-record prefix density*: $|S_{i+1}|/|S_i|$, the fraction of K_i-record orbits that achieve K_{i+1} as their next height record.

Numerically, the gate mass at $r_{K_{i+1}}$ is $\sim 40\%$ (for $K_i = 4$, $K_{i+1} = 5$), while the first-record prefix density is $\sim 15\%$—nearly three times smaller. The gate mass is not the relevant quantity for Theorem T; the prefix density is.

We prove **Lemma DSC-pre** (exact): after the K_i-cascade, the odd-core satisfies $q \equiv 3^{L_{K_i}} r_{K_i} \not\equiv r_{K_{i+1}} \pmod{2^{K_i}}$ (by Theorem UPI, Paper M). The orbit starts the inter-record period at a point that is *not* the K_{i+1} target; it must traverse the Syracuse renewal dynamics to reach $r_{K_{i+1}}$.

Conjecture DSC (numerical, open): $|S_{i+1}|/|S_i| \le 2^{-(K_{i+1}-K_i)}$ with $C < 1$ for all $K_i \ge 4$. Numerically $C \le 0.87$ in all measured cases, decreasing with K_i. This is strictly stronger than the HCL upper bound $C \le 1$, and if proved would yield $|S_\infty(N)| = 0$ without any mixing assumption.

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1 Introduction: The Gate Mass Trap

Papers P–U showed that TV mixing alone cannot prove the bound $|S_{i+1}|/|S_i| \le 2^{-(K_{i+1}-K_i)}$; the relevant decay is not about equidistribution of renewal residues, but about which specific orbits achieve a new height record. This paper separates the two quantities sharply.

2 Gate Mass vs First-Record Prefix Density

Definition 2.1 (Gate mass and prefix density)**.** Let $K_1 < K_2$.

- *Gate mass* $\gamma(K_1, K_2)$: among all gate-1 renewal events occurring after the K_1 record and before the first subsequent record of height $\ge K_2$, the fraction at which $q \equiv r_{K_2} \pmod{2^{K_2}}$.
- *Prefix density* $\rho(K_1, K_2)$: $|S_{K_1 \to K_2}(N)|/|S_{K_1}(N)|$, the fraction of K_1-record orbits that achieve K_2 as their immediate next height record.

Proposition 2.2 (They are different)**.** $\gamma(K_1, K_2) \ne \rho(K_1, K_2)$ *in general.*

Numerically ($N = 10^5$, $K_1 = 4$, $K_2 = 5$):

	Value	Context
Gate mass $\gamma(4,5)$	~ 0.39	Among 325,430 gate-1 renewals
Prefix density $\rho(4,5)$	0.1532	Among 43,620 orbits in $S_{(4)}$
HCL upper bound $2^{-(K_2-K_1)}$	0.5	Algebraic exact

The gate mass (39%) is the fraction of renewal *events* at the target residue. The prefix density (15%) is the fraction of *orbits* achieving the next record. These differ because:

1. Many orbits visit the r_{K_2} residue class at renewals that do not constitute a new K_2 record (because K_2 is not a new maximum, or the orbit converges before reaching K_2).
2. The first-record requirement selects only orbits where the *first* visit to r_{K_2} is at a strictly increasing height level.

3 The Starting Position After a Cascade

Lemma 3.1 (DSC-pre: off-target cascade start)**.** *After the K_i-cascade (cascade length $L_{K_i} = \lfloor (K_i - 2)/2 \rfloor$), the odd-core satisfies:*

$$q_{after} \equiv 3^{L_{K_i}} \cdot r_{K_i} \pmod{2^{K_i}}.$$

Moreover, $3^{L_{K_i}} \not\equiv 1 \pmod{2^{K_i}}$ *for all* $K_i \geq 4$ *(Theorem SR, Paper I). Therefore* $q_{after} \not\equiv r_{K_i} \pmod{2^{K_i}}$*, and in particular* $q_{after} \not\equiv r_{K_{i+1}} \pmod{2^{K_{i+1}}}$ *(since* $r_{K_{i+1}} \equiv r_{K_i} \pmod{2^{K_i}}$*).*

Proof. By Theorem OC (Paper G): one cascade step gives $q \mapsto 3q$, so L_{K_i} cascade steps give $q_{\text{after}} = 3^{L_{K_i}} q_{\tau_i} \equiv 3^{L_{K_i}} r_{K_i} \pmod{2^{K_i}}$. By Theorem SR (Paper I), $3^{L_{K_i}} \not\equiv 1 \pmod{2^{K_i}}$, hence $q_{\text{after}} \not\equiv r_{K_i}$. □

Remark 3.2 (Structural meaning)**.** Lemma DSC-pre says: the orbit starts its inter-record phase at a *specific, deterministic* value within the r_{K_i} class—but NOT at the K_{i+1} target $r_{K_{i+1}}$. The orbit must travel through renewal dynamics to reach $r_{K_{i+1}}$. This travel distance, not just the target density, determines the prefix density.

4 Numerical Evidence for Conjecture DSC

K_i	K_{i+1}	$\lvert S_{i+1}\rvert/\lvert S_i\rvert$	HCL $2^{-(K_{i+1}-K_i)}$	C
4	5	0.183	0.500	0.37
4	6	0.108	0.250	0.43
4	7	0.025	0.125	0.20
4	8	0.006	0.063	0.09
5	6	0.194	0.500	0.39
5	7	0.081	0.250	0.32
5	8	0.018	0.125	0.14
6	7	0.033	0.500	0.07
6	8	0.016	0.250	0.06

All measured $C < 1$; the strict inequality $|S_{i+1}|/|S_i| < 2^{-(K_{i+1}-K_i)}$ holds in every case. Moreover, C is decreasing as K_i grows.

Theorem 4.1 (FPO — First-Passage Obstruction, exact)**.** *For each* $K_i \geq 4$*, all orbits in* $S_i(N)$ *begin their inter-record phase from the* same *cascade endpoint modulo* $2^{K_{i+1}}$*:*

$$q_{\text{cascade}} \equiv 3^{L_{K_i}} \cdot r_{K_i} \pmod{2^{K_{i+1}}}, \quad q_{\text{cascade}} \not\equiv r_{K_{i+1}}.$$

Consequently, $|S_{i+1}(N)|/|S_i(N)|$ *equals the* first-passage probability *from the fixed start* q_{cascade} *to the fixed target* $r_{K_{i+1}}$ *under gate-1 renewal dynamics—a purely deterministic quantity, independent of any mixing assumption.*

Proof. By Theorem JE: $q_{\tau_i} \equiv r_{K_i} \pmod{2^{K_i}}$ for all orbits in S_i. After L_{K_i} cascade steps (Theorem OC): $q_{\text{cascade}} = 3^{L_{K_i}} q_{\tau_i}$, so $q_{\text{cascade}} \equiv 3^{L_{K_i}} r_{K_i} \pmod{2^{K_i}}$—a fixed value for all orbits in S_i. By Theorem SR (Paper I): $3^{L_{K_i}} \not\equiv 1 \pmod{2^{K_i}}$, so $q_{\text{cascade}} \not\equiv r_{K_i}$, hence (by HCL) $q_{\text{cascade}} \not\equiv r_{K_{i+1}} \pmod{2^{K_{i+1}}}$. □

Open Problem 4.2 (Conjecture DSC)**.** For all $K_i \geq 4$ and $K_{i+1} > K_i$:

$$\text{FPP}(K_i \to K_{i+1}) := \frac{|S_{i+1}(N)|}{|S_i(N)|} \leq 2^{-(K_{i+1}-K_i)}$$

with $C \leq C_0 < 1$. Proved rigorously, this closes Theorem T without any mixing assumption.

5 Proof Strategy for Conjecture DSC

Remark 5.1 (Why TV mixing fails here)**.** The approach of using Paper B's TV mixing theorem fails because:

1. The relevant quantity is a *first-passage* probability (first time $q \equiv r_{K_2}$), not a marginal distribution.

2. First-passage probabilities depend on the specific starting point (Lemma DSC-pre: the cascade endpoint $3^{L_{K_i}} r_{K_i}$) and the entire path, not just the limiting distribution.

3. TV convergence to Haar does not control first-passage time distributions; these require a different tool (e.g., spectral gap estimates for the Green's function, or direct path counting).

Remark 5.2 (Why Conjecture DSC is natural)**.** The inequality $|S_{i+1}|/|S_i| \leq 2^{-(K_{i+1}-K_i)}$ follows from HCL *if* each orbit in S_i visits all extensions of r_{K_i} equally. Conjecture DSC asserts this inequality holds even without uniform visitation—in fact more strongly ($C < 1$). The data suggests that orbits starting at $3^{L_{K_i}} r_{K_i}$ (off the K_{i+1} target) have *lower* than average probability of hitting $r_{K_{i+1}}$ before converging.

6 Summary

Result	Status	Section
Gate mass $\neq$ prefix density (distinction)	Clarification	§2
Lemma DSC-pre: cascade starts off-target	**Proved, exact**	§3
Conjecture DSC: $\lvert S_{i+1}\rvert/\lvert S_i\rvert \leq 2^{-(K_{i+1}-K_i)}$, $C < 1$	**Open (numerical)**	§4
Proof strategy: first-passage vs marginal	Direction	§5

References

[1] H. Kamanoi, "Conditioned Mixing Transfer" (Paper U), 2026.

[2] H. Kamanoi, "Hensel Conditional Scarcity" (Paper T), 2026.

[3] H. Kamanoi, "Shrinking Targets" (Paper I), 2026.

[4] H. Kamanoi, "Odd-Core Explosion" (Paper G), 2026.